

Dear Laura: Ethnographic Lettering as Critical Technical Practice in Design

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Abstract

Within HCI, ethnography has been embraced to grapple with the complexity of real-world settings and design technologies that operate effectively “in the wild.” However, some have criticized HCI’s engagement with ethnographic methods, arguing that ethnography has been circumscribed to an empirical, rather than analytical, enterprise whose perceived utility is limited to communicating uncomplicated facts to inform near-term design goals. We explore how “ethnographic lettering” might deepen HCI’s engagement with ethnography as a critical technical practice by allowing researchers to reframe technology users as interlocutors and the primary audience for the knowledge they produce. Ethnographic lettering can create avenues for more accountable and analytical reflections on the design of interactive systems. By experimenting with relational writing and research methods, we open the possibility for sociotechnical system design to be taken up not only as objects of study but as sites of critical inquiry and radical experimentation towards more just social formations.

CCS Concepts

• **Social and professional topics** → **Socio-technical systems**; • **Human-centered computing** → **Ethnographic studies**; • **Applied computing** → *Law, social and behavioral sciences*.

Keywords

lettering, critical data, ethnography, justice, carceral systems

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1 Introduction

Dear Laura,¹

A few months ago, I looked up the organizational chart for your jurisdiction’s Court Administration

¹Laura is a pseudonym.

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(CA) and I found a new, unfamiliar name written at the top of Pretrial Services. It took me by surprise — I never thought you’d actually leave the CA. But everyone has a breaking point. Maybe you finally reached yours. The last time I saw you, I could tell you were beginning to fray at the edges. We’d spent months collaborating on a project to build a “judge dashboard” that could surface trends in the ways judges made pretrial decisions like setting bail. It was an exciting project, one that we hoped might restore a sense of justice and fairness to the courts. But some judges viewed the work as a threat to their autonomy and authority and, before long, our project was put on an “indefinite pause.” During our last face-to-face meeting, a liaison from your boss’s office came to brainstorm some ideas for how we might better spend our time. But you already had your next data project underway. You pulled out your phone to show us a photo of four large stacks of paper, about a foot tall each.

Those small mountains of paperwork represented your jurisdiction’s in-custody jail population on a single day, around twelve thousand people. You planned to carry them with you to the state legislature in a couple of months as a way of illustrating the sheer scale of the problem you were dealing with.² In the meantime, four members of your staff had been assigned the monumental task of going through each page manually to check for errors in the data. That was a lot of manpower, especially since your department was understaffed. But you were willing to invest the time and resources in order to make your case.

“I try to dispel every single myth or reaction that the data is not good,” you explained, “Because that’s what they’ll say, ‘Well I don’t trust your data.’... Well, I have four staff members looking up all twelve thousand people, double checking their stats.... And if they don’t trust the data then we can have Trish and Bob and Lori and Trinity³ say, ‘Well, we did this. You don’t trust what we did?’”

It was such a classic move on your part, to try to persuade legislators of the dire need to decrease jail populations through brute force diligence. If only your data were cleaner, more accurate, then surely

²The number of legally innocent people incarcerated pretrial increased by 433 percent nationwide from 1970 to 2015, almost entirely due to increases in cash bail assignments (Preston and Eisenberg 2022).

³All of these names are pseudonyms.

it would get the point across. Surely, the numbers would speak for themselves. At the time, I pretty much agreed with this sentiment. Our collaboration was based on the premise that, if only we had the right kinds of data, the right quality assurance processes in place, then we could support more fair and humane decisions in the courts. But now I am wary of the many ways that expanded data collection fuels a culture of punishment, creating insurmountable mountains of moral debt for the people caught up in the system. I used to have faith in the power of well-reasoned, data-driven dialogue to shift the worldviews of people in positions of authority. But I no longer underestimate the power of fuzzy logic to fuel the specter of violence in the courts.

It was probably for the best that our project was shut down. Our work was constrained by powerful limitations, and, by accommodating those limitations, we ceded too much ground. Data is often framed as a pragmatic first step toward criminal justice reform, a place for establishing common ground in order to support more fair outcomes. I was initially drawn to our project because it seemed do-able. But these days I'm more interested in expanding what's possible through data than I am in settling for what's practical. We've never gotten the chance to talk about this kind of stuff. That's why I'm reaching out to you now. Because I know your commitment to bringing about a more just world is real. And I know that, in order to seed a better future, it is essential that we reflect on our failed experiments, that we learn from our mistakes.

2 Ethnographic Lettering As Critical Technical Practice

This article explores the ways that “ethnographic lettering” [31] might deepen HCI's engagement with ethnography as a critical technical practice [1]. Within critically-oriented design communities, HCI scholars have framed design as a knowledge enterprise that can shed light on the complex and contingent underpinnings of our social world [3, 4, 6, 40, 41]. To achieve this, the designers of socio-technical systems must expand their boundaries of inquiry beyond the technical considerations of a given technology to include the social, organizational, and cultural factors that govern their use [16, 19].

Within HCI, ethnography has been embraced as a means of grappling with the complexity of real world settings and designing technologies that operate effectively “in the wild.” However, some have criticized HCI's engagement with ethnographic methods, arguing that ethnography has been circumscribed to an empirical, rather than analytical, enterprise whose perceived utility is limited to the communication of relatively uncomplicated facts to inform near-term design goals [8, 9, 13].

Others have warned that ethnography limits, rather than expands, the engagement of users in the design process, as HCI workers maintain the “verbal privilege” [28, 33] of translating complex

social realities into design mandates that have immense impact on both the process and outcomes of design projects [13, 28]. These scholars encourage the HCI community to prioritize users, rather than designers, as the primary audience for their findings, arguing that such a shift opens up possibilities for more fundamental reformulations of design challenges, ones that are not tethered to a specific project or problem framing [14].

This article explores ways that “ethnographic lettering” [31] might address these concerns and deepen HCI's engagement with ethnographic practice. Laurence Ralph ([31], p. 194) describes ethnographic lettering as both a mode of writing and a research methodology that enables researchers to participate in “an ‘unfinished’ and ever evolving dialogue” with the people involved in the social problems they study. Ethnographic lettering is a relational practice that gives researchers the chance to reframe their research subjects as interlocutors and the primary audience for the knowledge they produce. It also enables researchers to build accountability into the claims they make in their work.

This article is written in an epistolary format, and is addressed to a former collaborator of the first author, who led a Pretrial Services Agency in a mid-size US state. As the resident social scientist on the project, the first author's work was initially situated as what Dourish and Button [15] term a “service science,” whereby she used qualitative research methods to inform the design and implementation of a “judge dashboard” that could surface trends in the ways judges responded to specific pretrial reforms. Of particular interest to the Court Administration was understanding the extent to which pretrial risk assessments were effective at “nudging” [36] judges to make less punitive decisions regarding bail.

By immersing herself as a courtroom insider, the first author had the opportunity to map out the contingencies and ambiguities that arose in the construction of criminal legal data, as well as the political and ideological struggles that shaped how that data was ultimately interpreted and used. This work complicated the view on a widespread theory of change within “data for social good” circles: namely, that we can use data to course correct the flawed habits of decision-makers in high-stakes realms, with the help of more accurate and objective information. It also helped us to appreciate the radical potential of this project, which flipped the script of most data-driven reform narratives by using data to model and intervene on the behaviors of judges, rather than defendants. By tracking the ways that judges set bail after the introduction of pretrial risk assessments, the first author and her collaborators were well positioned to complicate a popular narrative regarding data-driven reform in the courts, as well as build in novel forms of data-driven accountability for the ways judges made decisions.

The project was shut down before the team had a chance to complete it. While this was initially disappointing, over time, the first author has come to appreciate the many lessons learned in pursuing this failed project. This letter is her attempt to share those insights with her former collaborator, with whom she had not spoken for three years before writing this. The impetus for writing the letter stems from a growing interest in experimenting with more relational forms of ethnographic research, ones that might enable us to center specific relationships in our analysis and form arguments as part of an ongoing dialogue, rather than as a definitive endpoint to the discussion.

3 Dear Laura (continued)

We'd set out to solve a tough problem. Money bonds were originally introduced as an incentive mechanism for getting people to show up to court. But these days it's widely acknowledged that judges use money bond⁴ to by-pass rules and regulations that are intended to preserve the presumption of innocence and give every person their fair day in court. When we interviewed judges about their thought process for setting bail, most of them frankly described the way they used money bonds to detain people pretrial, rather than as a tool for incentivizing people to come to court. As one judge explained to me, "Ultimately, my decision is, do I want you out or do I want you in. If I want you out, I'm going to set a bond either ROR⁵ or something I think you can post. If I don't want you out, I'll set a bond you can't post".

I was surprised by how forthright judges were in talking about money bail in this way. I'd been thinking of the abuse of money bail as an open secret, something that everyone knew happened, but nobody talked about, especially in reference to their own work. But the practice has become so normalized, that judges had no qualms in talking about it. I think a big reason for this was because the use of financial bonds as a means of detention was framed as an essential tool for maintaining public safety. Throughout our interviews, judges repeatedly invoked fear of releasing someone who might go on to commit a "horrendous," "scary," or "tragic" crime. Take, for example, the following judge, who was widely known to as a pro-reform official who supported decreasing jail populations:

Judge D: I'm all for incarcerating people post-conviction, if that sentence is what is appropriate. I'm not okay with incarcerating people presumed innocent, unless we need to keep the community safe... The only people we should be holding (pretrial) are the really dangerous people.

On its face, this line of reasoning seems hard to argue with. However, there's a massive gap between this rhetoric and the reality of mass pretrial incarceration. Less than five percent of arrests in the United States are for crimes involving physical harm to another person [22]. In fact, the incidence of arrest for such crimes during the pretrial stage is so low that it is statistically impossible to meaningfully differentiate people according to their level of risk [12]. Given these statistical realities, scholars have argued in unequivocal terms that essentially no one poses a high enough public safety risk to justify jail time before they've been convicted

⁴A money bond, also known as bail, is a payment made to the court to secure a defendant's temporary release from jail while awaiting trial. The purpose of bail is to guarantee that the defendant will appear in court as required. If the defendant does not appear, the court keeps the money.

⁵ROR is short for "release on own recognizance," which means that a person is released from jail without paying a bond or being subject to other conditions for release.

of a crime [34]. Yet, judges in your jurisdiction continue to detain around a third of their pretrial population on any given day [23].

These incarceration rates are fueled by a psychological environment that makes pretrial detention feel necessary, even in the face of overwhelming evidence to the contrary. In theory, it should have been easy to dispel such concerns, especially for someone like you, who was well positioned to wield data to facilitate good-faith dialogues with judges. Through our judge interviews and surveys, we'd hoped to surface widely held beliefs and heuristics that could be tested against the CA's internal data. Statements like this:

Judge B: For a very serious offense, and even trafficking charges, which can be very serious, once we have them they're gonna have an incentive to leave. And I am always surprised when they do post bond and they do come back.

Such statements could be reformulated into testable hypotheses that served as a springboard for more in-depth conversations with judges. For example, is it true that people facing serious charges (i.e., felonies) are more likely to fail to appear in court than someone charged with a misdemeanor? We hoped to use the CA's historical data to answer such targeted questions, and in doing so, subtly challenge the beliefs and rules of thumb that propped up carceral outcomes in the courts. But we didn't fully appreciate how foundational court data was for sustaining the boogie man in the first place. We underestimated how hard it would be to invert the narrative lens used to interpret court data, away from evaluating the supposed risk of criminalized people and toward understanding the ways data are used to justify ongoing criminalization.

3.1 Creating Boogie Men Through Moral Debt and Fuzzy Logic

Over the course of our collaboration, I grew to appreciate how committed you were to using data to engage in earnest debates with judges regarding various bail reforms. For example, back in 2011, you polled all the judges across the jurisdiction, asking them for input on what factors they thought were most relevant for evaluating a person's pretrial risk. Then you systematically tested each of those factors to identify which ones should be included in your jurisdiction's validated risk assessment instrument. Through that process, you were able to narrow down the set of factors which had a significant statistical correlation with the specific pretrial outcomes that judges were supposed to consider.

In spite of these efforts, a number of judges expressed a lack of trust in the data used in your risk assessment. For example, on multiple occasions, judges

and court officials recounted a story of a case in Furrow County⁶ where a person had been brought into court with a prior escape charge, only to escape again by jumping through a courthouse window. As the story goes, the man leapt onto the courthouse roof and managed to cross the street before the police were able to rearrest him. When Pretrial administered his risk assessment, he came back as “low risk” for Failure to Appear (FTA). I don’t know if this story is true, but it’s the kind of tale that was used repeatedly to illustrate how ridiculous it was that prior escapes were not factored into FTA scores.

It seems intuitively correct that anyone who has tried to abscond from law enforcement or a detention facility would pose a higher risk for not showing up to court. However, as our team learned more about the evolution of this data point, we began to realize that the issue was much more complicated than it appeared at first glance. As it turned out, the category of escape did actually correlate with future failures to appear, from 1995 until around 2010. After that, however, prior escape charges stopped being predictive of any pretrial outcomes. This was largely due to the fact that home incarceration (HIC) had become a much more prevalent practice in your jurisdiction. As the jail overcrowding crisis ballooned into unsustainable proportions, more judges began to incarcerate people in their homes, where they were monitored using a GPS tracking device. Whenever the home incarceration device registered someone as stepping outside of their prescribed area of confinement, or stopped working for some reason, then that event was registered as an “escape” in the court data.

Home incarceration devices are notoriously finicky.⁷ In fact, false alerts are so prevalent that supervising officers in many jurisdictions often neglect to follow up when they are sent an alert [24]. Such technical difficulties make it challenging for people to meet their basic needs. For example, there have been multiple reports of people being fired from their jobs after taking breaks from work in order to walk around outside to reconnect their device to the network [25]. In your jurisdiction, many violations occurred when people momentarily stepped outside of their area of confinement to do things like grocery shopping. Such incidents are a classic example of the ways that technical violations are used to criminalize survival, by penalizing people for trying to meet their basic needs. They can hardly be equated with escaping from a jail or courthouse.

As home incarceration events became a larger proportion of the data points classified under “escape,” this input became less and less predictive of future pretrial outcomes. As a result, a researcher who’d been

hired to locally validate your tool recommended that it be omitted from the risk assessment. Yet, judges interpreted this as a willful omission of relevant facts and consistently listed it as a reason that they did not trust the scores offered up by the risk assessment.

As I learned more about these views, I began to see the ways that the data-fication of certain events had become a kind of demerit system, a way of penalizing pretrial defendants by giving them dings on their record that could be used to justify their ongoing punishment down the line. Prior to the launch of your validated risk assessment, you tried to minimize the prevalence of this practice. When your agency migrated its records over to a new database system, you did a full overhaul of your team’s data entry processes. A key goal of this work was to minimize the tendency for certain categories, like failure to appear and arrest, to become inflated by other events. For example, you implemented quality assurance processes in order to ensure that pretrial workers did not record a “failure to appear” on a proof of compliance event⁸ or an “arrest” on a failure to pay warrant. This kind of quality assurance was essential if you were to use administrative data to evaluate specific pretrial risks.

It also curtailed the ability for penal officials to use data as a way of assigning what I’ve come to understand as “moral debt” to incarcerated people, debt that is virtually impossible to repay. At its core, data collection within the criminal legal system is geared towards capturing negative interactions with law enforcement, such as prior arrests, charges and convictions, as well as minor technical violations like failure to pay a fine. There are very few opportunities for people to demonstrate through data the ways that they are improving their lives or contributing positively to society. Take for example, a story that Fred, a member of your team, told me, about a man who had been subjected to an extreme amount of mandatory drug testing while awaiting a resolution to his case. For months, the man was steadfast in his commitment to meeting his conditions of release. He showed up two to three times a week to pee in a cup and every single one of his drug tests came back negative. Yet, after submitting to hundreds of drug tests and making all of his court appearances, he missed his court date for sentencing. “We beat him down so much that he finally gave up,” Fred explained.

When the man did eventually return to court, the judge rebuked him. Rather than consider all the ways the man had demonstrated his willingness to comply with court conditions, the judge focused primarily on his one misstep. Such examples illustrate the ways that the “data as demerit” paradigm stabilizes the punitive nature of the criminal legal system. When

⁶Furrow County is a pseudonym.

⁷For example, in California, researchers found that the monitors used by the Department of Corrections failed their accuracy test in 46 of 102 tests [24].

⁸A proof of compliance event can include a wide range of activities, such as calling one’s pretrial supervisor or showing up to a clinic for drug testing.

a person fails to jump through even a minor bureaucratic hoop, that failure ends up on their permanent record, which can be dredged up again and again to reinforce a narrative that this person poses a “risk” to the community.

As different spheres of the criminal legal system become more integrated through shared databases, then it becomes even harder to counteract the accumulation of moral debt over time. Data becomes a shared structural resource that creates a contagion of cultural stigma across jurisdictions and institutions [38]. It is often impossible to challenge such “derogatory data” [18], which over time is used to justify the exclusion of people from their full legal rights. In this way, criminal legal data becomes not just an archive, but an arsenal [18].

Such practices are key to understanding how the courts manufacture moral debt through official record-keeping, and also how such data drive growing racial disparities in the system. Decades of research have shown that, for the same conduct, Black and Latinx people are more likely to be arrested, prosecuted, convicted and sentenced to harsher punishments than their white counterparts [11, 26, 32]. People of color are treated more harshly than similarly situated white people at every stage of the criminal legal system, which results in serious racial biases in the data. Unsurprisingly, when judges incorporate this discriminatory data into their decision-making processes, it produces discriminatory results. These discrepancies not only maintain, but expand existing disparities, by creating self-fulfilling prophecies through data that justify racial profiling and disparate treatment.

It was an uphill battle to fight against this demerit system, especially in the context of pretrial, because judges held strong views about what information was relevant for their decisions. Although you did your best not to inflate certain data categories on your pretrial risk assessment, your agency was obliged to include additional information in your pretrial reports, according to what a given judge thought was important. For example, in some jurisdictions your staff included information about whether or not an individual was on probation or had previously violated a condition of his release. Judges could then use this information however they saw fit, which often involved spinning up their own “sincere fictions” [38] about the ways such data implicated a person in risky behavior.

But this practice alone doesn’t account for the visceral responses that judges expressed when discussing the prospect of reducing pretrial jail populations. Judges need to interpret data through a lens that significantly raises the perceived stakes of such demerits, transforming minor technical violations into indicators of menacing future behavior. This was achieved through a process which I call “fuzzy logic,” where judges developed their own homespun narratives about

the relevance of certain data points in order to ratchet up the perceived risks of pretrial release. It was one of those things that judges returned to over and over when they discussed reasons why they strayed from the recommendations offered in the pretrial risk assessment.

For example, judges grossly over-weighted the importance of prior involvement with the system in their estimates of danger. When I asked one judge to explain how she evaluated the risk of pretrial violence, she said, “If they have a prior serious offense, such as assault, I would give them 75%. I would say the same thing if they’re currently on probation or parole. I would give all of those a great deal of weight.” These estimates of risk are completely divorced from reality. Nobody has a seventy-five percent chance of failing to appear or being arrested for a violent crime while awaiting trial. In fact, the judge’s estimates are more than ten times greater than the prevalence of arrest for a violent crime in the pretrial population. Yet such patterns of thought persist with the help of fuzzy logic, wherein judges manufacture their own explanations for the relevance of certain data points in order to magnify the perceived riskiness of a person facing trial.

As someone who believed in quality assurance processes, you held strong to the conviction that specific data points should be tightly coupled with the outcomes they are supposed to measure. However, there was one major exception to this policy — the inclusion of “New Criminal Activity” (NCA) scores on your agency’s pretrial risk assessment. The NCA score evaluated a person’s likelihood of being arrested for any crime, even though arrest for any crime is not a legally permissible reason to detain someone before their trial. Unsurprisingly, judges incorporated NCA scores into their evaluations of public safety risk, which led to a massive expansion in the perceived potential for pretrial violence. Take, for example, the following, where we asked a judge to reason through the use of NCA to evaluate public safety risk:

Chelsea: Should we be using general re-arrest data for nonviolent offenses to evaluate risk of dangerousness?

Judge B: Yes. I just think that generally all prior criminal history should be considered and all current charges. I don’t think we should put on blinders one way or the other whether they’re serious or not serious...damaging property...criminal mischief, first degree. That’s a felony for damaging property. Is it the kind where it’s stupid kids that are out spray painting the side of a building? Or is it gangsters out spray painting the side of the building that they are tagging?...is it a [domestic violence] kind of situation where they’ve destroyed a car and are torturing these

souls? That all sounds like a property offense.

As this judge argues, even seemingly minor charges, such as a property offense, could be masking the capacity for more serious violence in the future. However, studies have shown that the people with the highest risk of arrest for a violent crime are not the same people with the highest risk for being arrested for any crime at all [27]. As Mayson ([27], 562) argues, “If the goal is to avert violence, focusing on the likelihood of any future arrest targets the wrong individuals and fails to target the right ones.” In other words, whether or not a person goes on to be arrested for a non-violent offense is just not relevant. But judges have their own ways of reasoning through arrest data, which fuels a looming sense of danger.

In the above statement, the judge makes her point by distinguishing the behavior of “stupid kids” from “gangsters,” where the latter group is implicitly understood to be more dangerous and threatening. Such language is loaded with racial undertones that enable the judge to evoke a generalized state of anxiety and fear of racialized subjects — the “gangster,” the “thug,” the “bad hombre” — without ever explicitly bringing race into the conversation. Given the historical discrepancies in arrest rates across racial groups, this slippage between arrest and danger enables judges to maintain racist cognitive schemas that rationalize the uneven distribution of justice in the courts.

In contrast to other battles you were fighting, the conflation of pretrial danger with arrest was widely accepted and normalized in the CA. When speaking to your staff, arrest for any new charge was used interchangeably with danger. These views are broadly reinforced through academic and institutional narratives, which habitually conflate generalized arrest with public safety threat [10, 36, 39]. In doing so, your staff, and the broader milieu of experts working on bail reform, institutionalized fuzzy logic, transforming shaky leaps in reasoning into common sense wisdom. Such practices extend the power of the state to criminalize people, not only through laws, but through more symbolic processes that create categories of people deemed unworthy of protection under the law. In this way, racialized criminalization is able to thrive in the full light of day.

I’ve come to understand criminalization through data as a core competency of the criminal legal system, essential for maintaining its relentless pace of expansion over time. Far from providing an objective view on individual risk, official court records serve as a tool for manufacturing moral debt through a demerit system that often punishes people for simply trying to survive. With the help of fuzzy logic, court officials then use that moral debt to conjure a specter of violence that legitimates the punishment of legally

innocent people. When combined, data as moral debt and fuzzy logic conjure a boogie man that is hard to dispel from the minds of court officials. Even in your agency, where there was top-down support for using data in a different way, these old habits had a way of creeping back in.

3.2 What now?

Our collaboration gave me a healthy dose of skepticism regarding the role of data in supporting transformative change in the courts. Data is often framed as a pragmatic first step toward criminal justice reform, a place for establishing common ground in order to support more fair outcomes. I was initially drawn to this idea because it seemed practical, but these days I’m more interested in expanding the horizon of what’s considered possible. That’s because I no longer accept the common sense wisdom which has so profoundly circumscribed the “solution space” of bail reform for all these years. I wonder if you feel the same?

Whenever I start to speculate about what you’re up to now, my mind goes back to that photo of those four stacks of paper in your office. It’s clear to me now that, even back then, you were beginning to reckon with the limits of using data to internally transform the culture of the courts. That in-custody report represented a shift in strategy, away from nudging judges towards less punitive decisions, and toward building external support for limiting judges’ ability to wield carceral tools like cash bail. Data-based argumentation was still an integral part of your approach, but your audience was different. I think the impulse to change audiences was a good one. By then it was clear that the punitive culture of the courts was so deeply ingrained that it wasn’t going to change without clear outside pressure.

If we are going to restore justice to the courts, then we have to eliminate the loopholes that drive mass incarceration, like cash bail. Insiders like you are in a critical position to identify and advocate for the removal of these kinds of administrative alibis for pretrial punishment. A few years ago, this might have seemed like an unachievable goal. But advocates are already pushing the agenda of pretrial reform beyond what was once considered do-able. For example, in 2021, Illinois became the first state in the country to pass a bill that completely eliminated the use of cash bail. The bill was the culmination of several years of community organizing, in which local advocates worked to build a broad movement to support decarceration.

Data played an important role in achieving this goal, but not in the way one might think. Rather than build tools to forecast pretrial risk, community volunteers observed court proceedings and collected data about judge decisions. They then used that data to

fuel public accountability campaigns that pressured judges to follow the law. This work is based on the assumption that the core challenge of bail reform was not informational, but political in nature.

The process of data collection is a process of rendering some things visible and other things invisible. In the criminal legal system, data regarding the decisions and actions of key officials, such as judges, prosecutors, and police officers, are scant. Court watching initiatives like the one in Cook County aim to create new data sets which enable communities to build stronger systems of accountability around decision makers and ensure that reform efforts translate into real change. But they're also extremely labor intensive and challenging. Volunteers' requests for access to public data, such as docket information, are frequently denied without further explanation. At other times, government offices rescind access to data that was initially made public, after that data was used to catalyze social movements. As a result, some researchers and community groups have undertaken labor intensive data collection initiatives in order to record information that the courts already possess but will not provide.

Court insiders like yourself are in a powerful position to advocate for the removal of such barriers. Rather than expand data collection about the people targeted by the criminal legal system, you could improve access to basic information about how many people are behind bars. You could help shift the way we interpret court records, away from evaluating the supposed risk of criminalized people and toward building accountability for the officials who consistently stray from the letter of the law.

None of these propositions will be easy to achieve. But the good news is that you don't have to do it alone. There is a growing movement of people who are already reframing the debate with regard to bail reform, renegotiating the terms of engagement in productive ways. Such people are committed to experimenting towards a freer future beyond the mainstream solution space. The pursuit of such goals will inevitably involve frustration and failure, but even our struggles can serve as stepping stones to success, if only we can learn from our mistakes. Our time together helped me to see powerful ideas and institutions in a new light. It helped me think through political possibilities and strategies that, for me, were previously unthinkable. This letter is my humble attempt to share those learnings with you, to keep the conversation going.

Take care,
Chelsea

4 Ethnographic Lettering: Reflections on A Relational Research Practice

It took me a few weeks to finally muster the courage to call my collaborator once this letter was complete. I was unsure how she would receive the news that I'd divulged the nitty-gritty details of our collaboration in a public piece of writing. She is a competent and energetic bureaucrat, one who's spent decades working inside the administration of the courts. In contrast, I'd spent most of my time since our collaboration trying to create new avenues for external accountability for decision-makers within the legal system. This difference in approach was part of why I wanted to write to Laura in the first place. She is someone whose perspective I value and respect, even though it varies quite a bit from my own.

When I finally did reach out to her, she received my letter with an open mind and an immense amount of generosity. She spent nearly two hours on the phone with me, reviewing each line of my writing to ensure that it was as accurate as possible. To my relief, she didn't have any major points of contention with the way that I'd portrayed her or her agency. But she also didn't fully agree with all of my conclusions. Moving forward, we plan to have another conversation that probes more deeply into her takeaways from our collaboration. I don't know how this experiment will end, but I hope that it results in both of us expanding our capacity to envision and enact freer futures, through and beyond data.

The design of sociotechnical systems is not simply about the creation of software and hardware objects, but the negotiation of practices and possibilities for living life together differently [35]. Ethnographic lettering opens up the possibility for the design of sociotechnical systems to be taken up not only as objects of study, but as sites of critical inquiry and radical experimentation towards more just social formations. Within HCI, there is growing interest in the development of relationship-centered approaches to technology design which build our collective capacity to create emergent alternatives to existing social conditions [37]. In this work, joy is often a central organizing concept [21]. Joy is the growth of people's capacity to perceive the world more expansively and live in ways that were previously unimaginable [5]. Cultivating joy is a collective pursuit — it's about being in relationship with people who push us to perceive the world more expansively and hold us accountable to living in ways that manifest the latent potential for justice in our everyday lives [20]. Such a process often requires us to engage with the "ragged edges," or residual tensions and regrets, of a given project in order to surface insights that might otherwise remain hidden from view [29].

The above letter represents one humble experiment in cultivating a joyful engagement with a former collaborator of the first author's, by developing a set of "common notions" [5] about why our project failed, and why similar projects are likely to fail in the future. These insights have greatly shaped our perspective on the opportunities and limitations of pursuing transformative social change through data, particularly from within institutions with deeply embedded power asymmetries and cultures of authority. More generally, ethnographic lettering could provide an approach to ethical computation that de-centers the data in "data ethics" in favor of cultivating a relationship-based theory of change, building our collective capacity to experiment, learn, and build new

worlds together. This experiment reframes the residual tensions and frustrations of a failed collaboration instead as an opportunity to generate insights, using affect as a guide to surface tensions and insights that might otherwise remain hidden or unresolved. Otto et al. deem these ragged edges both ethnographically significant and generative in the design process [29]. This involves rigorously reflecting on what's worked and what hasn't in our own small experiments to build technologies that make the world a more livable place for everyone.

5 Deepening Ethnographic Practice in HCI

The most powerful engagements with ethnography in design are those which surface under-examined assumptions and produce insights whose relevance extend well beyond a given design prompt [17]. In this article, the first author explores the ways ethnographic lettering can deepen ethnographic practice in HCI, by creating avenues for more accountable and analytical reflections on the design of interactive systems. In the above letter, she outlines the ways that expanded data collection in the criminal legal system fuels a culture of punishment, creating insurmountable mountains of moral debt for the people caught up in the system. She then outlines the ways that judges use “fuzzy logic” to conjure a specter of violence that legitimates the punishment of legally innocent people. When combined, data as moral debt and fuzzy logic conjure a boogie man that is hard to dispel from the minds of court officials.

This discussion troubles a widely held theory of change in “data for social good” circles — namely, that algorithms and data can be used to course correct the cognitive pitfalls and implicit biases of key decision makers by presenting evidence-based claims about the likelihood of future events. By “studying up” [2] on judges and judicial culture, the above project upset current framings of both the challenges of bail reform and the ethical stakes of algorithmic systems in criminal justice. When designers flip the script on prevailing narratives in this way, they can reveal the often-hidden contours of the political economy underlying the design of data-driven systems. In doing so, we make way for alternative visions of “data for social good” to emerge, ones which shed light on the structural and cultural factors which perpetuate social inequality and lay the groundwork for more robust forms of accountability for people and institutions in positions of power and authority.

The above letter opens up important conversations regarding the ways that data more generally operate as a “negative credential” [7, 30] that justifies the ongoing marginalization of targeted groups. It also illustrates the ways that the users of data systems develop their own “sincere fictions” [38] about the relevance of certain data points in order to strengthen dominant systems of meaning and control. Such insights can't be surfaced through mathematical models or resolved through narrow technical efforts to make algorithms more “explainable” [16]. They require a more holistic engagement with the socio-political factors which shape the ways people make sense of algorithmic systems in a particular context. Future research in HCI could build from this work, by attending to the ways certain narratives are enabled and others are silenced by organizational and cultural factors which fall beyond the bounds of algorithmic formalism, through the process of constructing and interpreting contemporary data archives.

Ultimately, the goal of this article is to experiment with creative avenues for deepening the role of ethnography and other qualitative methods in the design of sociotechnical systems. The most powerful engagements with ethnography in design are analytical in nature, rather than empirical, and often involve a critical reflexive practice that surfaces under-examined assumptions and produces insights that extend well beyond a given design prompt. Ethnographic lettering offers an opportunity for designers to reinvigorate their qualitative fieldwork as a critical technical practice that can shed light on the unexamined political and relational challenges that so powerfully shape the work of technology design.

References

- [1] Philip E. Agre. 1997. Toward a Critical Technical Practice: Lessons Learned in Trying to Reform AI. In *Social Science, Technical Systems, and Cooperative Work: Beyond the Great Divide*. Psychology Press, 131–157.
- [2] Chelsea Barabas, Colin Doyle, JB Rubinovitz, and Karthik Dinakar. 2020. Studying up: reorienting the study of algorithmic fairness around issues of power. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. 167–176.
- [3] Jeffrey Bardzell, Shaowen Bardzell, and Lone Koefoed Hansen. 2015. Immodest proposals: Research through design and knowledge. In *Proceedings of the 33rd annual ACM conference on human factors in computing systems*. 2093–2102.
- [4] Ditte Amund Basballe and Kim Halskov. 2012. Dynamics of research through design. In *Proceedings of the Designing Interactive Systems Conference*. 58–67.
- [5] Carla Bergman and Nick Montgomery. 2017. *Joyful militancy: Building thriving resistance in toxic times*. AK Press.
- [6] John Bowers. 2012. The logic of annotated portfolios: communicating the value of ‘research through design’. In *Proceedings of the designing interactive systems conference*. 68–77.
- [7] Sarah Brayne. 2014. Surveillance and system avoidance: Criminal justice contact and institutional attachment. *American Sociological Review* 79, 3 (2014), 367–391.
- [8] Elizabeth F Churchill. 2017. Data, design, and ethnography. *interactions* 25, 1 (2017), 22–23.
- [9] Andrew Crabtree, Tom Rodden, Peter Tolmie, and Graham Button. 2009. Ethnography considered harmful. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (CHI '09). Association for Computing Machinery, New York, NY, USA, 879–888. doi:10.1145/1518701.1518835
- [10] Matthew DeMichele, Megan Comfort, Kelle Barrick, and Peter Baumgartner. 2021. The intuitive-override model: Nudging judges toward pretrial risk assessment instruments. *Fed. Probation* 85 (2021), 22.
- [11] Stephen Demuth and Darrell Steffensmeier. 2004. The impact of gender and race-ethnicity in the pretrial release process. *Social Problems* 51, 2 (2004), 222–242.
- [12] Karthik Dinakar, Chelsea Barabas, and Colin Doyle. 2019. Technical flaws of pretrial risk assessments raise grave concerns. <https://www.media.mit.edu/posts/algorithmic-risk-assessment/>
- [13] Paul Dourish. 2006. Implications for design. In *Proceedings of the SIGCHI conference on Human Factors in computing systems*. 541–550.
- [14] Paul Dourish. 2007. Responsibilities and implications: further thoughts on ethnography and design. In *Proceedings of the 2007 conference on Designing for User eXperiences*. 2–16.
- [15] Paul Dourish and Graham Button. 1998. On “technomethodology”: Foundational relationships between ethnomethodology and system design. *Human-computer interaction* 13, 4 (1998), 395–432.
- [16] Upol Ehsan, Q Vera Liao, Michael Muller, Mark O Riedl, and Justin D Weisz. 2021. Expanding explainability: Towards social transparency in ai systems. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–19.
- [17] Virginia Eubanks. 2018. *Automating inequality: How high-tech tools profile, police, and punish the poor*. St. Martin's Press.
- [18] Community Justice Exchange. 2022. From Data Criminalization to Prison Abolition. <https://www.communityjusticeexchange.org/en/resources-all/from-data-criminalization-to-prison-abolition>
- [19] Ben Green and Salomé Viljoen. 2020. Algorithmic realism: expanding the boundaries of algorithmic thought. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*. 19–31.
- [20] Alexis Pauline Gumbs. 2014. Prophecy in the present tense: Harriet Tubman, the Combahee pilgrimage, and dreams coming true. *Meridians* 12, 2 (2014), 142–152.
- [21] Alexis Hope, Catherine D'Ignazio, Josephine Hoy, Rebecca Michelson, Jennifer Roberts, Kate Krontiris, and Ethan Zuckerman. 2019. Hackathons as participatory design: Iterating feminist utopias. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14.

- [22] Community Resource Hub. 2019. The Vera Institute of Justice - Arrest trends. <https://communityresourcehub.org/resources/the-vera-institute-of-justice-arrest-trends/>
- [23] The Vera Institute. 2019. Incarceration Trends in Kentucky. <https://www.vera.org/downloads/pdfdownloads/state-incarceration-trends-kentucky.pdf>
- [24] Jack Karsten and Darrell M. West. 2017. Decades later, electronic monitoring of offenders is still prone to failure. <https://www.brookings.edu/blog/techtank/2017/09/21/decades-later-electronic-monitoring-of-offenders-is-still-prone-to-failure/>
- [25] James Kilgore, Emmett Sanders, and Kate Weisburd. 2021. The Case Against E-carceration. <https://inquest.org/the-case-against-e-carceration/>
- [26] Marc Mauer. 2010. Justice for all-challenging racial disparities in the criminal justice system. *Hum. Rts.* 37 (2010), 14.
- [27] Sandra G Mayson. 2017. Dangerous defendants. *Yale LJ* 127 (2017), 490.
- [28] Michael Muller. 2004. HCI as translation work: How translation studies can inform HCI research and practice. In *CHI Workshop on Reflexive HCI*.
- [29] Ton Otto, Jennifer Deger, and George E Marcus. 2021. Ethnography and exhibition design: Insights from the Moesgaard inaugural. *Design Studies* 74 (2021), 100989.
- [30] Devah Pager. 2008. *Marked: Race, crime, and finding work in an era of mass incarceration*. University of Chicago Press.
- [31] Laurence Ralph. 2020. *The torture letters: reckoning with police violence*. University of Chicago Press.
- [32] M Marit Rehavi and Sonja B Starr. 2014. Racial disparity in federal criminal sentences. *Journal of Political Economy* 122, 6 (2014), 1320–1354.
- [33] Adrienne Rich. 1993. *Your native land, your life*. WW Norton & Company.
- [34] Megan Stevenson and Sandra Mayson. 2021. Virtually No One is Dangerous Enough to Justify Jail. <https://theappeal.org/virtually-no-one-is-dangerous-enough-to-justify-jail/>
- [35] Lucille Alice Suchman. 2007. *Human-machine reconfigurations: Plans and situated actions*. Cambridge university press.
- [36] Cass R Sunstein. 2019. Algorithms, correcting biases. *Social Research: An International Quarterly* 86, 2 (2019), 499–511.
- [37] Alexandra To, Angela D. R. Smith, Dilruba Showkat, Adinawa Adjagbodjou, and Christina Harrington. 2023. Flourishing in the everyday: Moving beyond damage-centered design in HCI for BIPOC communities. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 917–933.
- [38] Nicole Gonzalez Van Cleve. 2022. Due process & the theater of racial degradation: the evolving notion of pretrial punishment in the criminal courts. *Daedalus* 151, 1 (2022), 135–152.
- [39] Arnold Ventures. 2017. Public Safety Assessment: A Risk Tool That Promotes Safety, Equity, and Justice. <https://www.arnoldventures.org/stories/public-safety-assessment-risk-tool-promotes-safety-equity-justice>
- [40] Tracee Vetting Wolf, Jennifer A Rode, Jeremy Sussman, and Wendy A Kellogg. 2006. Dispelling "design" as the black art of CHI. In *Proceedings of the SIGCHI conference on Human Factors in computing systems*. 521–530.
- [41] John Zimmerman, Erik Stolterman, and Jodi Forlizzi. 2010. An analysis and critique of Research through Design: towards a formalization of a research approach. In *proceedings of the 8th ACM conference on designing interactive systems*. 310–319.